

5(a)	<i>any one from :</i> <i>(when the air pressure lowers inside bronchus)</i> holds, bronchus / airway open ; A supports, bronchus / airway ; prevents, bronchus / airway, from collapsing ;	1
5(a)(ii)	<i>idea that</i> (cartilage arranged as) incomplete / C-shaped, rings in the trachea ; <i>idea of</i> more regular distribution (of incomplete rings) along the length of trachea ;	1
5(a)(iii)	<i>max 1 if cilia or mucus linked to incorrect cell</i> (ciliated epithelial cells have) cilia, to, waft / sweep / AW, mucus, away / up the trachea / AW ; (goblet cells that) secrete mucus to trap, dust / bacteria / pathogens ;	2
5(b)(i)	the larger elephant has a smaller surface area to volume ratio ; ora	1
5(b)(ii)	<i>any three from:</i> are multicellular / have many cells ; need a large quantity of / enough, oxygen to supply, every cell / all parts of body ; active so high demand for, oxygen / removal of CO₂ ; diffusion across body surface is not fast enough (to meet needs) ; AW long diffusion distance (so diffusion is not fast enough to meet needs) ;	3
5(b)(iii)	<i>any four from:</i> three polypeptide chains ; (three polypeptides) form triple helix ; A coil around each other R ref. to alpha helix hydrogen bonds hold the three. polypeptides / strands. together ; the three polypeptides, lie close together / form a tight coil / are tightly wound ; every third amino acid (in the polypeptide chain) is glycine / many glycine-proline alanine repeats / many gly-pro-ala repeats ; glycine is found on the inside of each, polypeptide / strand; AVP ; e.g. detail hydrogen bonds, between peptide bond NH (of a glycine) and C=O group in adjacent polypeptide	4

Question	Answer	Marks
5(b)(iv)	<i>allow reference to collagen fibrils instead of collagen molecules, in correct context (collagen molecules form fibrils, which form fibres)</i> <i>any one from:</i> <i>(many) collagen molecules lie parallel ;</i> <i>no weak points because</i> <i>strong / covalent, (cross) links between molecules ;</i> <i>(ends of) molecules (in, fibril / fibre), are staggered ;</i>	1

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